Haoyao Ruan, PhD Candidate

Available summer 2027 • TN-eligible (USMCA Statistician)

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Summary

Statistician with 3.5+ years in industry (2+ in drug development: Novartis, Cytel) and a PhD due summer 2027. Background spans **clinical trials and real-world evidence**, with hands-on execution of statistical analyses for oncology regulatory and HTA submissions. Experienced in **Bayesian methods**, **survival analysis** and indirect treatment comparisons (MAIC, STC). First author, ***JRSS Series A*** (2026). Proficient in R (8+ yrs) and SAS; CDISC ADaM datasets.

Experience

Summer 2026

Biostatistician III (Intern)
Novartis, Advanced Quantitative Sciences • East Hanover, NJ, USA

- Built an **R Shiny app** to monitor **survival data maturity** across oncology trials, enabling the clinical team to assess event-accrual readiness and inform analysis timing decisions.
- Modelled future event timing with a **piecewise exponential model**; outputs supported study execution team planning for a late-phase oncology programme.

Sep 2021 – Aug 2023

Research Consultant, Statistics (Full-time)
Cytel Inc., Real-World Evidence and Advanced Analytics • Toronto, ON, Canada

- Executed **statistical analyses and contributed to SAP development** for indirect treatment comparisons (MAIC, STC) supporting oncology **regulatory and HTA submissions** in relapsed/refractory multiple myeloma (*Adv Ther*, 2022) and BRCA1/2+ metastatic castration-resistant prostate cancer (*Adv Ther*, 2024).
- Co-authored a study evaluating **Bayesian borrowing** to augment an RWD-based **external control arm** in 1L NSCLC, examining replication of RCT survival estimates and BB's role as a precision and bias-assessment tool (*J Comp Eff Res*, 2024).
- First author on the variance–bias trade-off of covariate adjustment in **synthetic control methods** (*Value in Health*, 2022).

Oct 2019 – May 2021

Statistical Consultant (Full-time)
Environmental Resources Management (ERM) • Toronto, ON, Canada

- Built **multivariate models and statistical tests** for environmental and health risk assessments (BACI designs, GAMMs, bootstrap kernel density estimates); automated reporting pipelines in R Markdown/Shiny, cutting turnaround time by **60%**.

Education

2023 – 2027

Ph.D., Statistics and Data Science (expected summer 2027)
University of Texas at Arlington • Arlington, TX, USA

- Dissertation: *Bayesian modeling for partially observed and aggregate data*.
- Teaching Assistant: Bayesian Data Analysis (graduate), UTA.

2017 – 2019

M.Sc., Statistics
Simon Fraser University • Burnaby, BC, Canada

- Thesis: *Covariance-adjusted sparse reduced-rank regression for imaging-genetics data*.

2013 – 2017

B.Sc., Applied Statistics, with Distinction
University of Toronto • Toronto, ON, Canada

Skills

Methods

Bayesian hierarchical models, custom MCMC samplers in R (incl. Gibbs); indirect treatment comparisons (MAIC, STC); external and synthetic control arms; survival analysis; GLMMs; sparse reduced-rank regression; bootstrap; Monte Carlo simulation and power analysis

Skills (continued)

ML	Lasso, random forests, gradient boosting, neural networks, PCA, clustering, cross-validation
Data	Analysis using CDISC ADaM datasets
Software	R (8+ years; Shiny, R Markdown), SAS, Python, SQL, LaTeX, Git, HPC (TACC, Compute Canada)

Publications

Methods (First Author)

1. **Ruan**, **H.**, Wang, K., Lim, J., & Wang, X. (2026). Bias meets Bayes: Bayesian aggregation for peer and self ranking [Accepted]. *Journal of the Royal Statistical Society Series A*.

Drug Development and Comparative Effectiveness

1. De Santis, M., Breijo, S. M., Robinson, P., Capone, C., Pascoe, K., Van Sanden, S., Hashim, M., Trevisan, M., Daly, C., Reitsma, F., van Beekhuizen, S., **Ruan**, **H.**, Heeg, B., & Verzoni, E. (2024). Feasibility of indirect treatment comparisons between niraparib plus abiraterone acetate and other first-line poly ADP-ribose polymerase inhibitor treatment regimens for patients with BRCA1/2 mutation-positive metastatic castration-resistant prostate cancer. *Advances in Therapy*, *41*(8), 3039–3058.
2. Strüebing, A., McKibbon, C., **Ruan**, **H.**, Mackay, E., Dennis, N., Velummailum, R., He, P., Tanaka, Y., Xiong, Y., Springford, A., & Rosenlund, M. (2024). Augmenting external control arms using Bayesian borrowing: A case study in first-line non-small cell lung cancer. *Journal of Comparative Effectiveness Research*, *13*(5), e230175.
3. He, J., Berringer, H., Heeg, B., **Ruan**, **H.**, Kampfenkel, T., Dwarakanathan, H., et al. (2022). Indirect treatment comparison of daratumumab, pomalidomide, and dexamethasone versus standard of care in patients with difficult-to-treat relapsed/refractory multiple myeloma. *Advances in Therapy*, *39*(9), 4230–4249.

Conference Abstracts

1. Strüebing, A., McKibbon, C., **Ruan**, **H.**, Mackay, E., Dennis, N., Velummailum, R., et al. (2023). CO90 Augmenting Synthetic Control Arms Using Bayesian Borrowing: A Case Study in First-Line Non-Small Cell Lung Cancer. *Value in Health*, *26*(12), S30.
2. **Ruan**, **H.**, Springford, A., Gupta, A., & Mackay, E. (2022). MSR102 Variance-Bias Trade-Off in Covariate Adjustment in the Context of Synthetic Control Methods. *Value in Health*, *25*(12), S369.

In Preparation

1. **Ruan**, **H.**, et al. (in preparation). *Bayesian binomial decomposition for contaminated counts, with application to oral reading fluency assessment*.
2. **Ruan**, **H.**, et al. (in preparation). *A Bayesian predictive index for survival-data analysis readiness*.

Statistical Collaborations in Linguistics

Speech Communication (2025); *Int J Speech Technol* (2023, two papers); *Speech Prosody* (2026, two papers).

Service & Awards

2025 – Present	**President**, National Student Data Corps UTA Chapter; **Treasurer and Co-founder**, ASA Student Chapter at UTA.
2025, 2026	**Third Place** (twice), ASA North Texas Chapter student paper presentations.
2019	**Best Graduate Poster Presentation**, Canadian Statistics Student Conference.